
W0
T0
W1
Conv3x3
Conv1x1
Relu
Conv3x3
Add
Relu
T1
DLT
DLT
Matmul
Matmul
OffsetReduce
Relu
Conv3x3
Add
Relu
DLT+Concat
Matmul
Split
OffsetReduce
+Relu
Conv3x3
+Add+Relu
(i) Original
(ii) Intra-expression derivation
(iii) Inter-expression derivation
(a) Optimizations found by EINNET
c
h
w
r
s
Conv
(i) Conv
DLT
Split weight along r and s
(ii) Split weight
Matmul
OffsetReduce
(iii) Transform Conv to Matmul with OffsetReduce
(b) Optimization of Convolution found by EINNET
(i) Conv
DLT
OffsetConcat
(ii) Duplicate input
Matmul
(iii) Transform Conv to Matmul using im2col algorithm
(c) Im2col Optimization for Convolution